

Coverage Without Citizenship: California’s Medi-Cal Adult Expansion at 1-138 vs. 1-200% FPL

Abstract

Background: California’s Ages 26 through 49 Adult Full-Scope Medi-Cal Expansion, effective January 1, 2024, extended full-benefit Medi-Cal to income-eligible residents regardless of immigration status. Public-data evaluations of these state-funded immigrant coverage expansions are scarce because the American Community Survey (ACS) identifies noncitizens rather than undocumented residents and because the income-eligibility band that defines “Medi-Cal-relevant” is itself a researcher choice.

Methods: This study assembled a 2018-2024 IPUMS ACS state-year panel of low-income noncitizens ages 26-49 and estimated ridge-augmented synthetic-control (ASCM) gaps for California against a 43-state donor pool that excludes states with comparable adult immigrant-coverage policies. Two co-primary FPL bands are reported: 1-200% FPL (broad low-income noncitizen target cell) and 1-138% FPL (Medicaid-relevant comparison). Robustness includes a full FPL-band stress test (1-100, 1-138, 139-200, 1-200), small-cell donor sensitivity (minimum state-year ACS n in $\{10, 15, 25, 50\}$), and full leave-one-donor-out across all 43 donor states.

Results: Uninsurance is robustly lower in California across all FPL bands: the ASCM gap is -7.5 pp at 1-200% FPL, -11.1 pp at 1-138% FPL, -12.1 pp at 1-100% FPL, and -7.6 pp at 139-200% FPL. The Medicaid result is fragile to the FPL band. The Medi-Cal / means-tested public coverage gap is +11.8 pp at 1-200% FPL but flips to -4.1 pp at the Medicaid-relevant 1-138% FPL band; any-public flips from +12.1 to -2.4 pp on the same comparison. Full leave-one-donor-out within 1-200% holds the Medi-Cal gap in $[+9.6, +14.1]$ pp. Small-cell donor sensitivity is stable. The placebo-in-space uninsured rank statistic is $p = 0.341$.

Conclusions: California’s adult Medi-Cal expansion is associated with a directionally robust reduction in measured uninsured among low-income noncitizens ages 26-49 across multiple FPL bands. The headline “higher Medi-Cal coverage” claim is NOT robust: it survives at 1-200% FPL but reverses sign at the 1-138% Medicaid-relevant band. This paper is presented as a cautious public-data screen, not as a send-ready causal estimate.

Introduction

States have increasingly used state funds to close eligibility gaps left by federal Medicaid rules for noncitizens. California’s Ages 26 through 49 Adult Full-Scope Medi-Cal Expansion is a discrete eligibility change for a clearly defined low-income noncitizen target population. The policy question is not only whether

state-funded expansions increase enrollment, but whether they measurably reduce uninsurance and shift the composition of coverage in population survey data that are available quickly enough for policy evaluation.

This paper asks whether California’s Ages 26 through 49 Adult Full-Scope Medi-Cal Expansion, effective January 1, 2024, changed coverage among low-income noncitizens ages 26-49. The analysis is deliberately framed around the public-data estimand: ACS noncitizens in the exposed age and income cell. That target does not perfectly identify undocumented residents, but it captures the population group most likely to be moved by the policy.

A core finding of this version is that the headline Medi-Cal-gain claim is sensitive to a single researcher choice — the FPL band that defines the target cell. The paper reports both 1-200% FPL (the broad public-data target) and 1-138% FPL (closer to the Medi-Cal income screen) as co-primary specifications.

Methods

Data and Population

The analysis uses the IPUMS ACS 1-year extract for 2018-2024. The sample retains household residents who are noncitizens (`CITIZEN = 3`), have family income strictly greater than 0% and less than or equal to the FPL upper bound, and fall in the exposed age band: ages 26-49. Person-level records are collapsed to state-year cells using ACS person weights. The primary outcomes are uninsurance, Medicaid or means-tested public coverage, any public coverage, and private coverage. Insurance variables are harmonized from the staged IPUMS extract.

Policy Timing

The treatment is California’s Ages 26 through 49 Adult Full-Scope Medi-Cal Expansion, effective January 1, 2024. The primary post-treatment window is 2024. The donor pool excludes states with comparable adult immigrant-coverage policies during or near the analysis window: CO, DC, IL, MN, NY, OR, WA. The audited sensitivity pool re-includes MN because its adult MinnesotaCare expansion begins in 2025.

Co-primary FPL bands

This version reports two co-primary specifications: 1-200% FPL (broad low-income noncitizen target cell) and 1-138% FPL (Medicaid-relevant comparison). Two additional bands (1-100% and 139-200% FPL) are reported as sensitivity.

Statistical Analysis

For each outcome, the analysis estimates a state-level synthetic-control model and a ridge-augmented synthetic-control (ASCM) model. The donor weights

minimize pre-treatment differences between California and a convex combination of donor states. The augmented model uses ridge regression on donor pre-treatment histories to partially correct interpolation bias when California is not exactly in the donor convex hull. The tables report treated observed values, ASCM counterfactuals, gaps in percentage points, pre-treatment RMSPE, and a placebo-in-space rank statistic for uninsurance.

The estimator-comparison sensitivity also reports custom diagnostic approximations: a donor-mean DiD baseline, an SCM-DiD diagnostic (SCM post counterfactual adjusted by the average pre-period SCM gap), a ridge-regularized two-way FE imputation, and a low-rank interactive-FE diagnostic. These are NOT the formal package implementations of Arkhangelsky et al. (2021) synthetic difference-in-differences, Xu (2017) generalized synthetic control, or Athey et al. (2021) matrix completion. They are sensitivity diagnostics only.

Results

Headline

At 1-200% FPL the ASCM gaps are -7.5 pp (uninsurance), +11.8 pp (Medi-Cal / means-tested public), +12.1 pp (any public), and -2.8 pp (private). At 1-138% FPL the same gaps are -11.1 pp, -4.1 pp, -2.4 pp, and -2.1 pp.

Uninsurance is robustly negative across all four FPL bands tested. The Medi-Cal and any-public results are NOT robust to the FPL band; the broader 1-200% band shows a positive gap and the narrower 1-138% band shows a small negative gap. The private-coverage decline is small and stable across bands.

Within-band robustness

Within 1-200% FPL, the headline Medi-Cal gap is stable to leave-one-donor-out across all 43 donor states (range [+9.6, +14.1] pp; median +11.8 pp) and to small-cell donor sensitivity (range across minimum state-year n thresholds: [+9.8, +11.8] pp). The headline uninsurance gap is stable to leave-one-donor-out (range [-8.3, -6.3] pp) and small-cell screens.

Inference caveats

The placebo-in-space uninsurance rank statistic is $p = 0.341$, which is weak. Several placebo units obtain near-zero pre-RMSPEs, producing very large post/pre ratios that suggest overfitting risk from a 6-pre-year/43-donor configuration. There is one post year. None of these limitations is solved by the estimator-comparison diagnostics, which only check direction.

Discussion

The uninsurance reduction in California 2024 among low-income noncitizens ages 26-49 is the directionally robust finding of this screen. It survives at 1-

138% FPL, at 1-100% FPL, at 139-200% FPL, at 1-200% FPL, across leave-one-donor-out, and across small-cell donor sensitivity. The estimator-comparison diagnostics agree in direction.

The Medi-Cal-coverage gain is not robust. It depends on the FPL band: a positive gain at 1-200% FPL flips to a small negative gap at 1-138% FPL. The likely mechanism is composition: the 1-200% target cell includes 138-200% FPL noncitizens, where the California pre-2024 share with Medi-Cal-like coverage was below the comparison donor average, and where the 2024 jump may reflect a level shift in pre-existing program participation rather than the new adult expansion. A Medicaid-relevant specification (1-138% FPL) does not show the same jump. The paper does not adjudicate this; it reports both bands.

The policy implications are real but limited. State-funded adult immigrant coverage expansions appear capable of moving measured uninsurance in survey data, which matters for states considering expansions or retrenchments. However, the analysis does not identify undocumented residents directly, has one post year, uses approximate diagnostics rather than formal SDID or GSynth packages, and shows a weak placebo rank.

Several limitations are important. First, ACS citizenship does not distinguish undocumented immigrants from lawful permanent residents, visa holders, refugees, or other noncitizen groups. Second, the post-period is short. Third, donor-pool contamination is possible because immigrant coverage policies are changing quickly across states. Fourth, exact or near-exact pre-fit on several outcomes can reflect good matching or fragility from a short pre-period relative to the donor count. Fifth, the FPL-band sensitivity is itself a structural feature of public-data screens for Medicaid-like programs and should be treated as a primary disclosure rather than a footnote. Sixth, the upstream raw ACS rebuild from the 557 MB fixed-width source was not completed during the 2026-05-15 audit, so the inherited clean parquet is the auditable source; a full byte-for-byte raw-to-clean rebuild remains a future check.

Conclusion

California’s Ages 26 through 49 Adult Full-Scope Medi-Cal Expansion is associated, in this public-data screen, with lower uninsurance among low-income noncitizens ages 26-49. The Medi-Cal coverage gain depends on the FPL band: positive at 1-200% FPL, slightly negative at the Medicaid-relevant 1-138% FPL band. The current version is a public-data screen, not a send-ready causal claim, and should be presented as a working paper while the inferential layer (longer post-period, formal SDID/GSynth/MC packages, additional placebo strategies) is built out.

Sources

- California DHCS Adult Expansion: <https://www.dhcs.ca.gov/services/medical/eligibility/Pages/Adult-Expansion.aspx>
 - IPUMS ACS source extract documentation: [papers/noncitizen-coverage/data/scripts/01_extract](#)
 - Specification registry: [analysis/specification-registry.yml](#)
 - FPL-band sensitivity: [analysis/robustness/fpl_band_sensitivity.md](#)
 - Small-cell donor sensitivity: [analysis/robustness/small_cell_sensitivity.md](#)
 - Full leave-one-donor-out: [analysis/robustness/full_leave_one_out.md](#)
 - Audit triggering this version: [review/full-pipeline-audit-2026-05-15.md](#)
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Appendix

Appendix

Appendix Figure A1. Raw ACS Pre-Trends

See [data/output/pretrend_acs_coverage.png](#).

Appendix Figures A2-A5. Treated and Synthetic Paths

See [analysis/figures/synth_path_uninsured.png](#), [analysis/figures/synth_path_medicaid.png](#), [analysis/figures/synth_path_any_public.png](#), and [analysis/figures/synth_path_private_coverage.png](#).

Appendix Figures A6-A9. Gap Paths

See [analysis/figures/gap_path_uninsured.png](#), [analysis/figures/gap_path_medicaid.png](#), [analysis/figures/gap_path_any_public.png](#), and [analysis/figures/gap_path_private_coverage.png](#).

Appendix Table A1. Donor Weights

See [analysis/tables/table3_top_donor_weights.md](#) and [analysis/tables/donor_weights_by_outcome.md](#).

Appendix Table A2. Placebo-in-Space Diagnostics

See [analysis/robustness/placebo_in_space_uninsured.csv](#).

Table 1. Target-Population Coverage Summary

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Group	Period	Years	Weighted popula- tion	Uninsured (%)	Medicaid (%)	Any public (%)	Private (%)
California	Pre- period	2018, 2019, 2020, 2021, 2022, 2023	961,502	35.8	42.2	42.8	23.1
California	Post- period	2024	895,554	26.8	54.9	55.5	20.3
Primary donor pool	Pre- period	2018, 2019, 2020, 2021, 2022, 2023	63,958	49.7	17.5	18.1	34
Primary donor pool	Post- period	2024	69,525	47.6	18.2	18.9	34.9

Notes: Cells are state-year means from ACS person-weighted state-year aggregates.

Table 2. Main ASCM Coverage Estimates (1-200% FPL primary)

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Outcome	Post years	Observed (%)	ASCM counterfac- tual (%)	Gap (pp)	Pre- RMSPE (pp)	Placebo rank p
Uninsured	2024	26.8	34.3	-7.5	0	0.341
Any public coverage	2024	55.5	43.5	12.1	0.85	
Medicaid / means-tested public coverage	2024	54.9	43.1	11.8	0.49	
Private coverage	2024	20.3	23.1	-2.8	0	

Notes: Gaps are treated-state observed minus ridge-augmented synthetic-control counterfactual. Positive gaps indicate higher coverage; negative gaps indicate lower uninsurance or lower private coverage.

Appendix Table. FPL-Band Sensitivity

FPL-Band Sensitivity for the CA Medi-Cal 26-49 ASCM Screen

Each cell is the ridge-augmented synthetic-control (ASCM) gap in percentage points (treated minus synthetic counterfactual) for CA in the 2024 post-year, holding the donor pool fixed (excludes CO, DC, IL, MN, NY, OR, WA).

	Any public coverage	Medicaid / means-tested public coverage	Private coverage	Uninsured
('1-100% FPL', 2637)	11.9	10.46	-2.63	-12.13
('1-138% FPL (Medicaid-like)', 4202)	-2.37	-4.08	-2.12	-11.14
('1-200% FPL (current primary)', 6869)	12.06	11.76	-2.8	-7.46
('139-200% FPL', 2667)	15.34	14.32	-5.12	-7.6

Key finding

The Medicaid and any-public gaps are NOT robust to the FPL band. At the broader 1-200% FPL band (the current primary specification) the Medicaid gap is positive (+11.8 pp). At the more Medicaid-relevant 1-138% FPL band the same gap flips to negative (-4.1 pp). Uninsurance remains directionally negative across all four bands. The headline should report both bands honestly and demote the single 1-200% point estimate from a causal claim to a public-data screen.

Donor and cell-size notes

The minimum donor state-year ACS cell n varies by band; small cells are a known weakness (Vermont/Alaska/South Dakota). See `small_cell_sensitivity.{csv,md}` for a complementary stress test.

Appendix Table. Small-Cell Donor Sensitivity

Small-Cell Donor Sensitivity (1-200% FPL primary)

Each row drops donor STATES that ever have a state-year ACS cell with unweighted n below the minimum threshold. The ASCM gap at 2024 is re-estimated. The current primary uses no minimum (`min_donor_cell_n=0`).

	Any public coverage	Medicaid / means-tested public coverage	Private coverage	Uninsured
(0, 43)	12.06	11.76	-2.8	-7.46
(10, 41)	10.14	9.81	-2.82	-6.31
(15, 36)	10.7	10.34	-2.7	-7.29
(25, 34)	11.11	10.55	-2.72	-8.18
(50, 31)	10.64	10.39	-3.7	-7.41

Interpretation

If gaps move materially as the threshold tightens, the headline is driven by small donor cells (Vermont/Alaska/South Dakota concern from the 2026-05-15 audit). If gaps are stable, small donor cells are not the source of the result, and the FPL-band sensitivity remains the binding concern instead.

Appendix Table. Full Leave-One-Donor-Out

Full Leave-One-Donor-Out Sensitivity (1-200% FPL primary)

Drops each donor state once and refits the ASCM. Summary across all 43 donor states:

outcome_label	baseline	gap p10	gap p25	gap median	gap p75	gap max	opp
Uninsured	-7.46	-8.29	-7.57	-7.46	-7.39	-6.25	43
Any public coverage	12.06	10.45	12.06	12.06	12.06	15.81	43
Medicaid / means-tested public coverage	11.76	9.56	11.76	11.76	11.76	14.08	43

14.1 | 0.14088 | Custom diagnostic: ridge on state + year dummies + three covariates. NOT a formal untreated-potential-outcome imputation. | | Any public coverage | any_public | 2024 | Low-rank IFE diagnostic (approx.) | 55.5 | 43.4 | 12.1 | 0.120933 | Custom diagnostic: rank-2 SVD/ridge screen on donor panel. NOT the Xu (2017) generalized synthetic-control package. | | Medicaid / means-tested public coverage | medicaid | 2024 | Donor mean DiD | 54.9 | 42.9 | 12 | 0.119702 | Unweighted donor-pool mean change from the pre-period to post-year. | | Medicaid / means-tested public coverage | medicaid | 2024 | SCM-DiD diagnostic (approx.) | 54.9 | 43.1 | 11.8 | 0.117523 | Custom diagnostic: SCM post counterfactual plus the average pre-period SCM gap. NOT the Arkhangelsky et al. (2021) SDID estimator. | | Medicaid / means-tested public coverage | medicaid | 2024 | SCM | 54.9 | 43.1 | 11.8 | 0.117849 | Canonical nonnegative donor weights that sum to one. | | Medicaid / means-tested public coverage | medicaid | 2024 | ASCM | 54.9 | 43.1 | 11.8 | 0.117576 | Ridge-augmented synthetic control; primary estimator. | | Medicaid / means-tested public coverage | medicaid | 2024 | Two-way FE ridge imputation (approx.) | 54.9 | 40.8 | 14.1 | 0.141041 | Custom diagnostic: ridge on state + year dummies + three covariates. NOT a formal untreated-potential-outcome imputation. | | Medicaid / means-tested public coverage | medicaid | 2024 | Low-rank IFE diagnostic (approx.) | 54.9 | 42.9 | 12 | 0.119533 | Custom diagnostic: rank-2 SVD/ridge screen on donor panel. NOT the Xu (2017) generalized synthetic-control package. | | Private coverage | private_coverage | 2024 | Donor mean DiD | 20.3 | 24 | -3.7 | -0.0372468 | Unweighted donor-pool mean change from the pre-period to post-year. | | Private coverage | private_coverage | 2024 | SCM-DiD diagnostic (approx.) | 20.3 | 23.1 | -2.8 | -0.0279976 | Custom diagnostic: SCM post counterfactual plus the average pre-period SCM gap. NOT the Arkhangelsky et al. (2021) SDID estimator. | | Private coverage | private_coverage | 2024 | SCM | 20.3 | 23.1 | -2.8 | -0.0279975 | Canonical nonnegative donor weights that sum to one. | | Private coverage | private_coverage | 2024 | ASCM | 20.3 | 23.1 | -2.8 | -0.0279975 | Ridge-augmented synthetic control; primary estimator. | | Private coverage | private_coverage | 2024 | Two-way FE ridge imputation (approx.) | 20.3 | 25.3 | -5 | -0.050045 | Custom diagnostic: ridge on state + year dummies + three covariates. NOT a formal untreated-potential-outcome imputation. | | Private coverage | private_coverage | 2024 | Low-rank IFE diagnostic (approx.) | 20.3 | 22.4 | -2.1 | -0.0206841 | Custom diagnostic: rank-2 SVD/ridge screen on donor panel. NOT the Xu (2017) generalized synthetic-control package. | Notes: Positive gaps indicate higher coverage than the counterfactual; negative gaps indicate lower uninsured or lower private coverage. Rows labeled '(approx.)' are diagnostic approximations only.

Appendix Table. Leave-One-Donor-Out (high weight)

Leave-One-Donor-Out Sensitivity

Outcome	Dropped donor	Original donor weight (%)	Baseline gap (pp)	Leave-one-out gap (pp)	Change from baseline (pp)	N donors
Uninsured	MA	15.9	-7.5	-7	0.5	42
Uninsured	HI	14.2	-7.5	-8.2	-0.8	42
Uninsured	RI	7.8	-7.5	-6.3	1.1	42
Uninsured	VT	4.4	-7.5	-6.2	1.2	42
Uninsured	OH	4.3	-7.5	-7.1	0.4	42
Uninsured	ND	4.1	-7.5	-7.8	-0.4	42
Medicaid / means-tested public coverage	MA	60.2	11.8	14.1	2.3	42
Medicaid / means-tested public coverage	NM	20.2	11.8	9.8	-1.9	42
Medicaid / means-tested public coverage	SD	13.9	11.8	9.7	-2.1	42
Medicaid / means-tested public coverage	AK	2.3	11.8	11.8	0.1	42
Medicaid / means-tested public coverage	VT	2.2	11.8	9.6	-2.2	42
Medicaid / means-tested public coverage	DE	1.1	11.8	11.6	-0.1	42

Notes: Rows drop each high-weight donor from the primary donor pool and rerun ASCM for the first post year.

Appendix Table. Donor-Pool Sensitivity

Donor-Pool Sensitivity

Outcome	Post year	Donor pool	Excluded states	N donors	Observed (%)	ASCM counter-factual (%)	Gap (pp)
Uninsured	2024	Primary conservative pool	CO, DC, IL, MN, NY, OR, WA	43	26.8	34.3	-7.5
Uninsured	2024	Audited adult-policy-through-2024 pool	CO, DC, IL, NY, OR, WA	44	26.8	34.8	-7.9
Medicaid / means-tested public coverage	2024	Primary conservative pool	CO, DC, IL, MN, NY, OR, WA	43	54.9	43.1	11.8
Medicaid / means-tested public coverage	2024	Audited adult-policy-through-2024 pool	CO, DC, IL, NY, OR, WA	44	54.9	40.6	14.3

Notes: The audited adult-policy-through-2024 pool excludes states identified by KFF as having adult coverage regardless of immigration status by the analysis window and re-includes Minnesota, whose adult policy begins after 2024.

Appendix Table. Donor-Policy Audit

Donor-Policy Audit - California Medi-Cal 26-49 Expansion

Updated on 2026-04-27.

Decision Rule

The primary donor pool excludes states with adult immigrant-coverage policies that could directly affect the low-income noncitizen adult target cell during or near the ACS 2018-2024 window. The audited sensitivity pool follows the KFF March 2024 adult-coverage list and re-includes Minnesota because its adult MinnesotaCare expansion begins after the 2024 ACS post year. Child-only and pregnancy-only immigrant coverage policies are documented but are not treated as adult target-cell contamination.

Minnesota is excluded in the conservative primary pool because of near-window policy change, but the audited sensitivity pool re-includes it because the adult expansion begins in 2025.

Audit Table

	State	Primary	Adult-policy audit	Source
	State name	role		
CA	California	Treated state	Adult coverage regardless of immigration status phased in 2020, 2022, and 2024.	KFF 2025; California DHCS
CO	Colorado	Excluded from primary donor pool	State-funded OmniSalud/marketplace subsidies for adults regardless of immigration status.	KFF 2024; KFF 2025
DC	District of Columbia	Excluded from primary donor pool	Longstanding locally funded Healthcare Alliance coverage for low-income adults.	KFF 2024; KFF 2025
IL	Illinois	Excluded from primary donor pool	State-funded adult immigrant coverage for older adults and ages 42-64 during the window.	KFF 2024; KFF 2025
NY	New York	Excluded from primary donor pool	State-funded coverage for adults ages 65 and older regardless of immigration status beginning in 2023.	KFF 2025; NILC 2026
OR	Oregon	Excluded from primary donor pool	Healthier Oregon extended full OHP benefits regardless of immigration status to all income-eligible adults in July 2023.	KFF 2025; Oregon OHA
WA	Washington	Excluded from primary donor pool	Marketplace subsidies and July 2024 Apple Health Expansion create adult coverage-policy overlap.	KFF 2024; KFF 2025
MN	Minnesota	Excluded from primary donor pool	Adult MinnesotaCare expansion begins in January 2025, after the ACS 2018-2024 analysis window.	KFF 2025; Minnesota DHS
CT	Connecticut	Retained if otherwise supported	Children/pregnancy-focused state-funded coverage; no broad adult coverage in the analysis window.	KFF 2024; NILC 2026
ME	Maine	Retained if otherwise supported	Children/pregnancy-focused state-funded coverage; no broad adult coverage in the analysis window.	KFF 2024; NILC 2026

State	Primary	Adult-policy audit	Source
State name	role		
MA	Massachusetts	Retained if otherwise supported	Children/pregnancy-focused state-funded coverage; no broad adult coverage in the analysis window. KFF 2024; NILC 2026
NJ	New Jersey	Retained if otherwise supported	Children/pregnancy-focused state-funded coverage; no broad adult coverage in the analysis window. KFF 2024; NILC 2026
RI	Rhode Island	Retained if otherwise supported	Children/pregnancy-focused state-funded coverage; no broad adult coverage in the analysis window. KFF 2024; NILC 2026
UT	Utah	Retained if otherwise supported	Children-focused state-funded coverage; no broad adult coverage in the analysis window. KFF 2024; NILC 2026
VT	Vermont	Retained if otherwise supported	Children/pregnancy-focused state-funded coverage; no broad adult coverage in the analysis window. KFF 2024; NILC 2026

Sources

- KFF. More States Are Providing Fully State-Funded Health Coverage to Some Individuals Regardless of Immigration Status. May 1, 2024.
- KFF. State Health Coverage for Immigrants and Implications for Health Coverage and Care. May 29, 2025; updated September 12, 2025.
- National Immigration Law Center. Medical Assistance Programs for Immigrants in Various States. March 30, 2026.
- California Department of Health Care Services. Ages 26 through 49 Adult Full Scope Medi-Cal Expansion. <https://www.dhcs.ca.gov/services/medical/eligibility/Pages/Adult-Expansion.aspx>